LABORATORY WORK 4

Database / PostgreSQL

Report on Completed Tasks

This report documents the SQL queries and final results for all 15 tasks. The screenshots are presented in the order in which the work was completed.

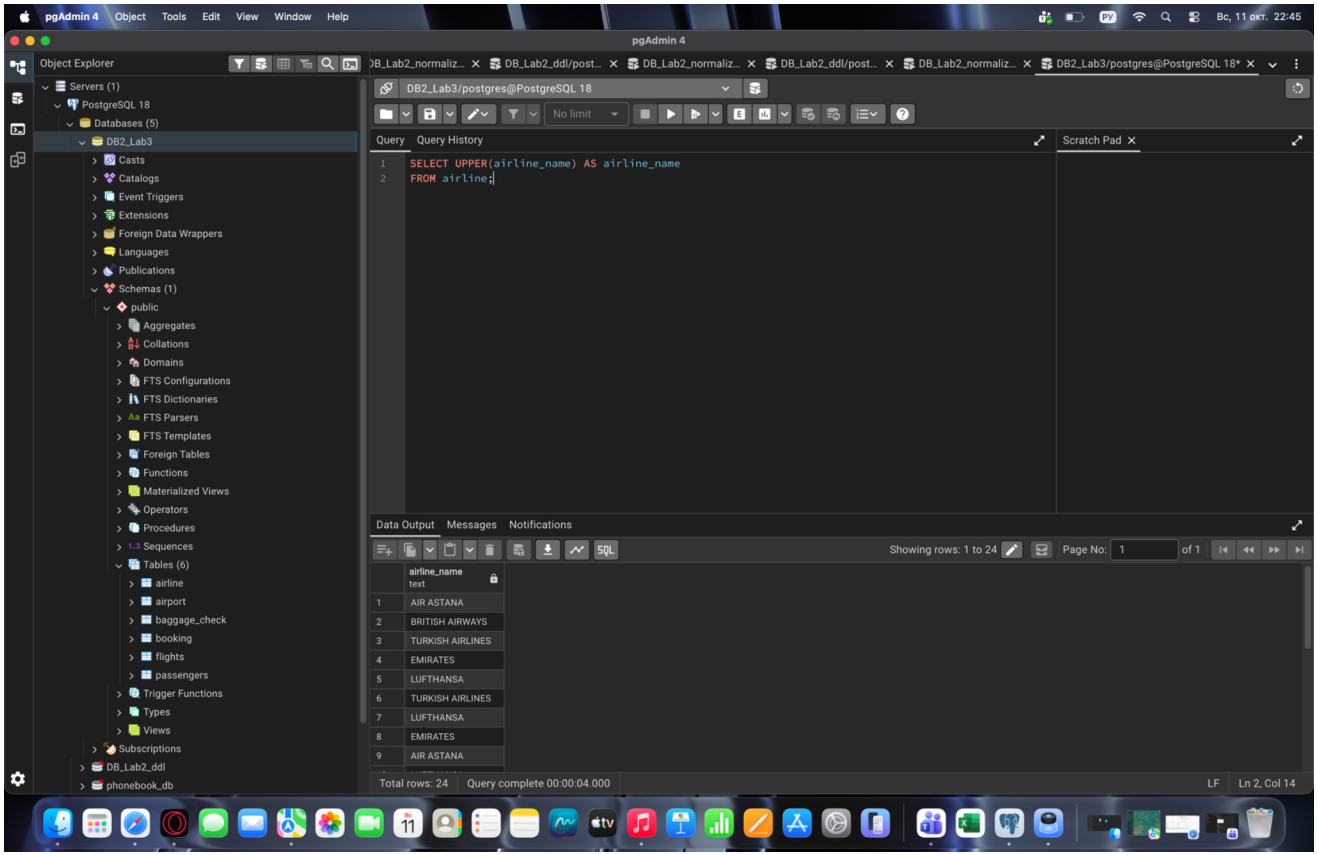
Task 1

Objective: Retrieve all airline names in uppercase.

SQL:

```
SELECT UPPER(airline_name) AS airline_name
FROM airline;
```

Result: The query returned 24 airline names in uppercase.



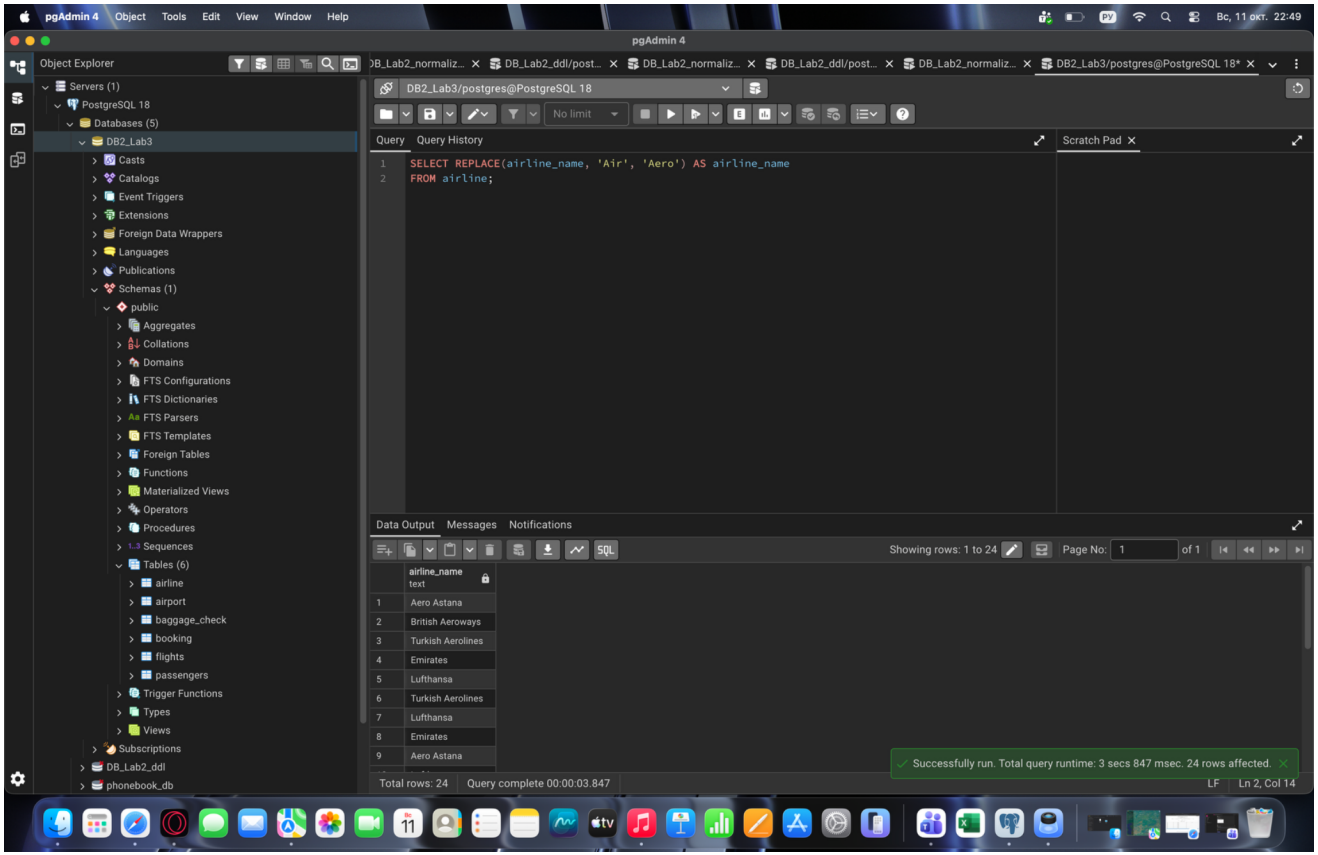
Task 2

Objective: Replace every occurrence of the word "Air" in airline names with "Aero".

SQL:

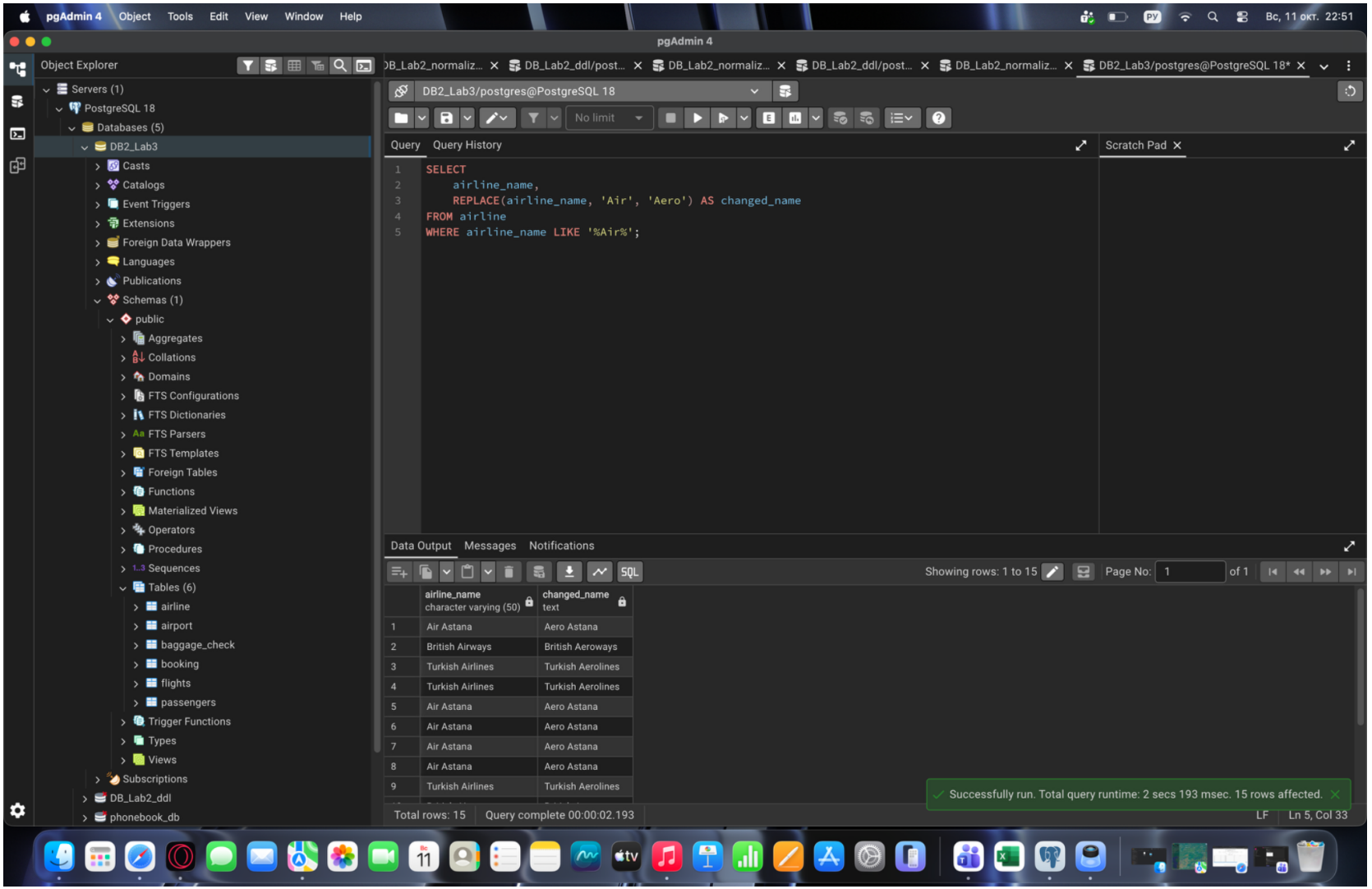
```
SELECT REPLACE(airline_name, 'Air', 'Aero') AS airline_name
FROM airline;
```

Result: The transformed airline names were returned. An additional query shows the original and changed values.



Task 2 - Result Verification

This additional screenshot shows verification of the query result.



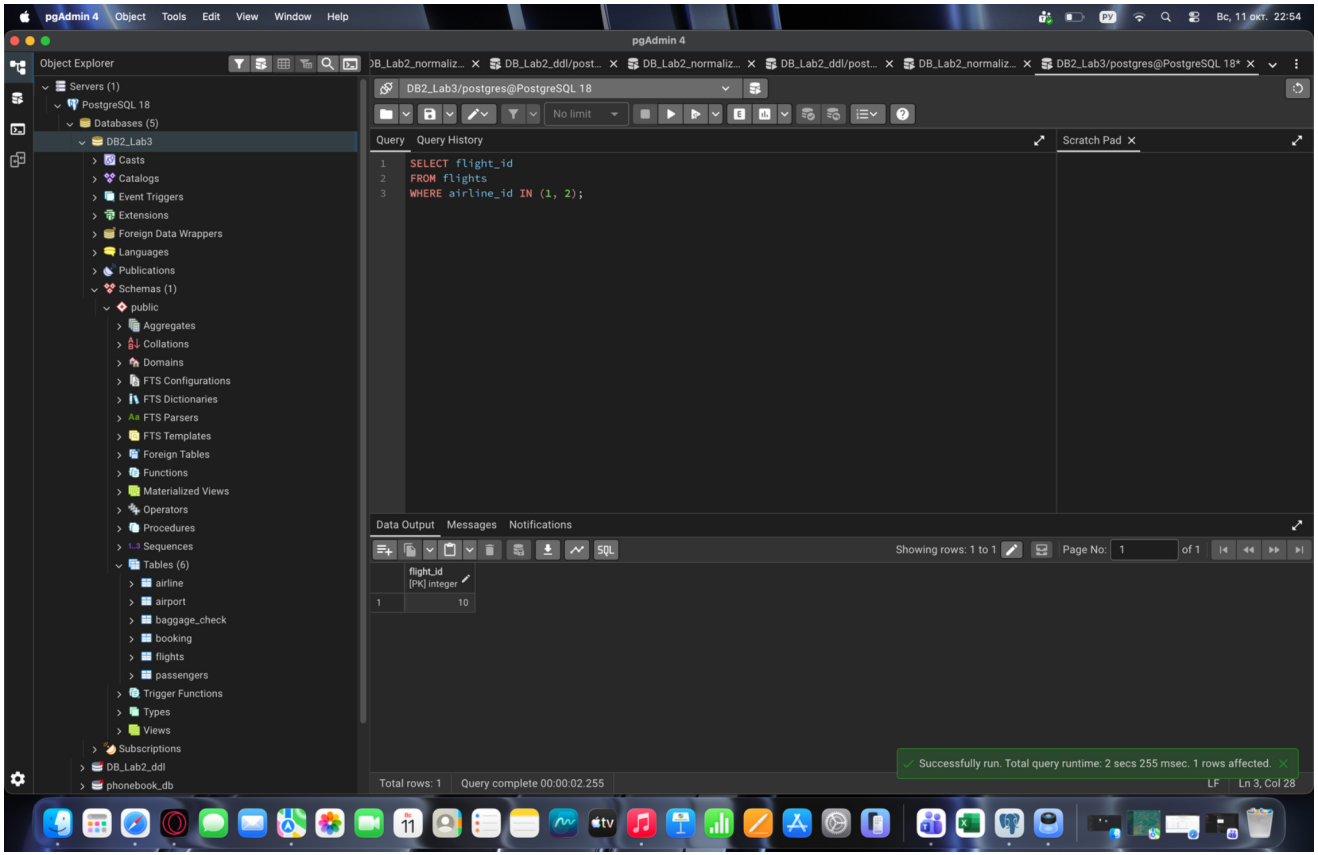
Task 3

Objective: Find flight numbers associated with airline 1 and airline 2. Since the current table has no flight_number column, flight_id was used.

SQL:

```
SELECT flight_id
FROM flights
WHERE airline_id IN (1, 2);
```

Result: The matching flight was found: flight_id = 10.



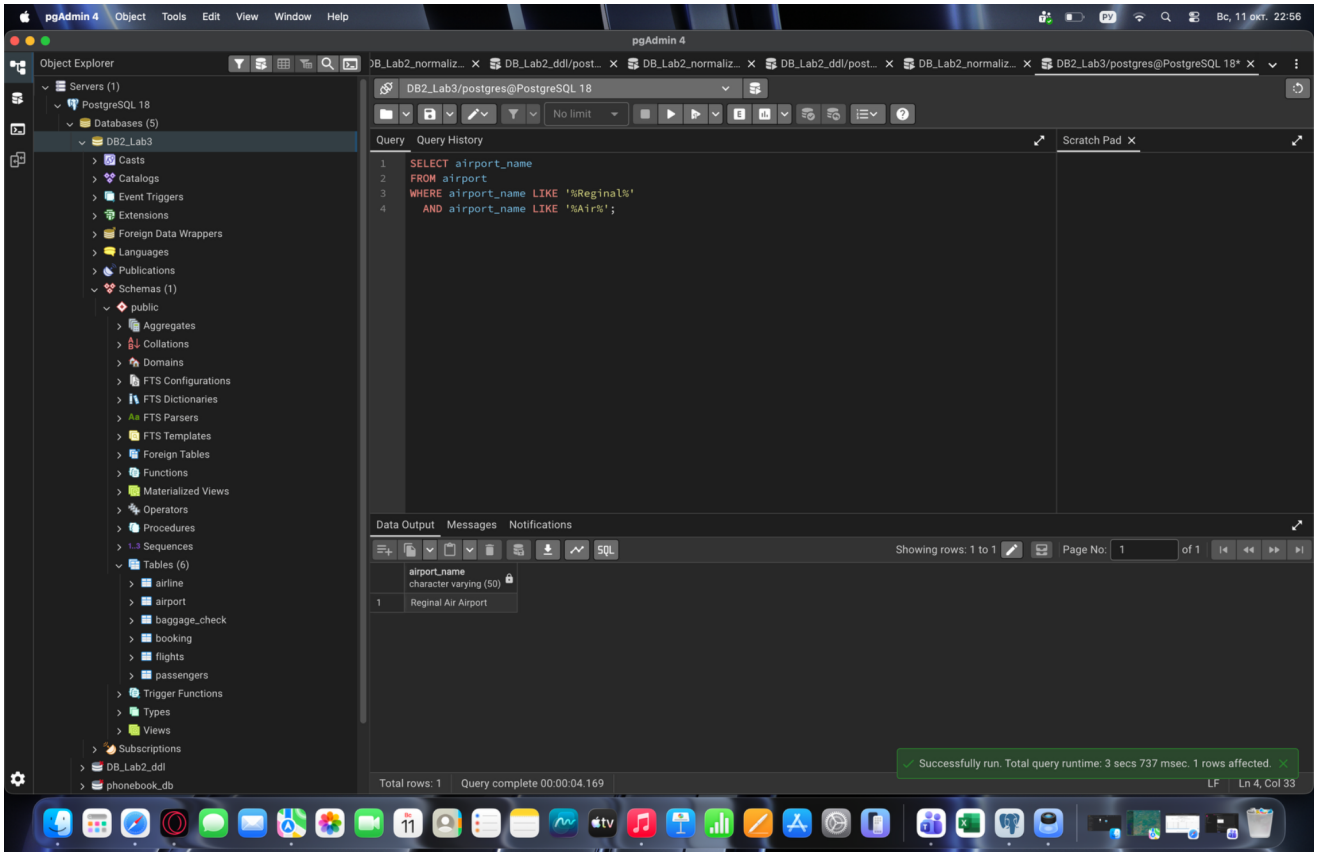
Task 4

Objective: Retrieve airports whose names contain both "Reginal" and "Air".

SQL:

```
SELECT airport_name
FROM airport
WHERE airport_name LIKE '%Reginal%'
  AND airport_name LIKE '%Air%';
```

Result: The matching airport "Reginal Air Airport" was returned.



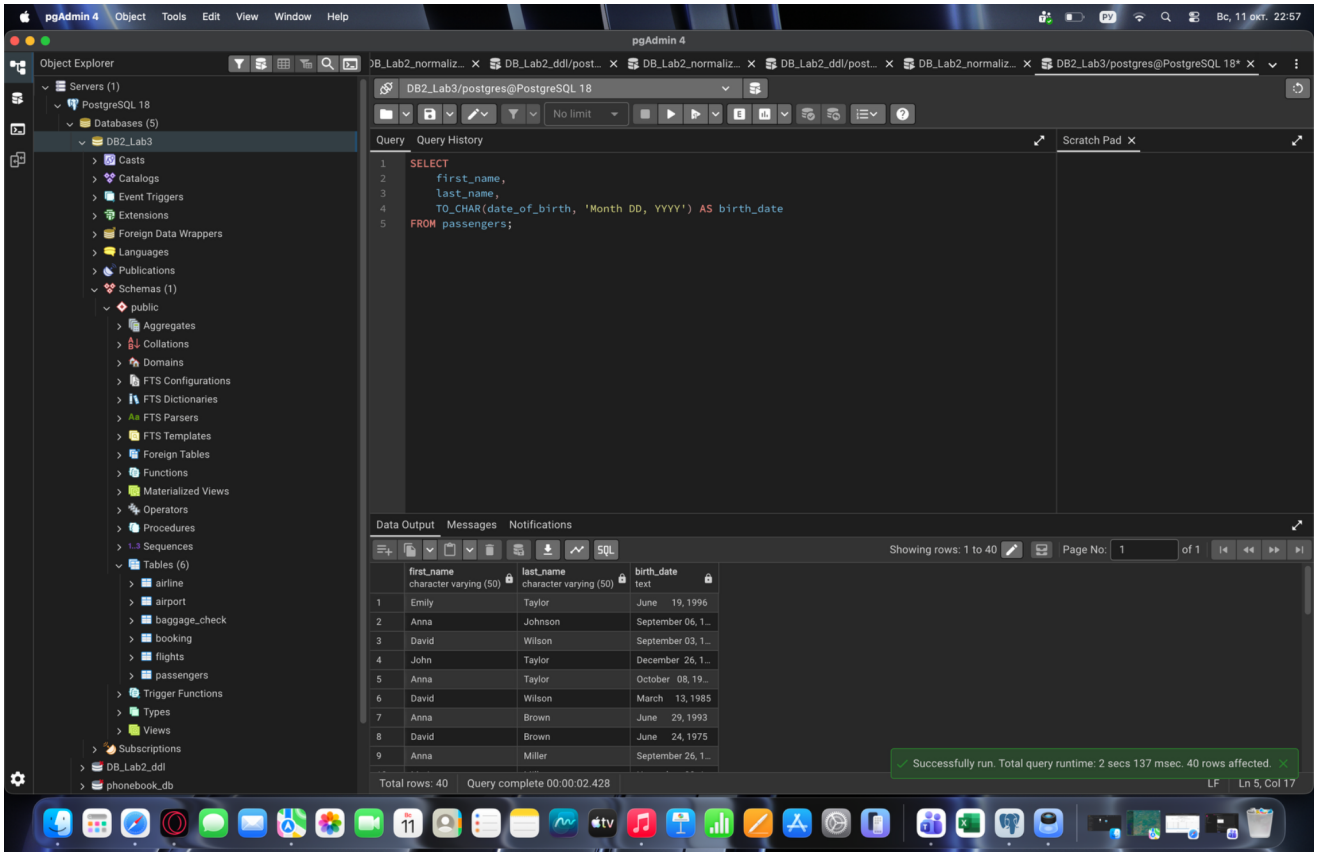
Task 5

Objective: Retrieve passenger names and format birth dates as Month DD, YYYY.

SQL:

```
SELECT
    first_name,
    last_name,
    TO_CHAR(date_of_birth, 'Month DD, YYYY') AS birth_date
FROM passengers;
```

Result: 40 passengers were returned with birth dates in the required format.



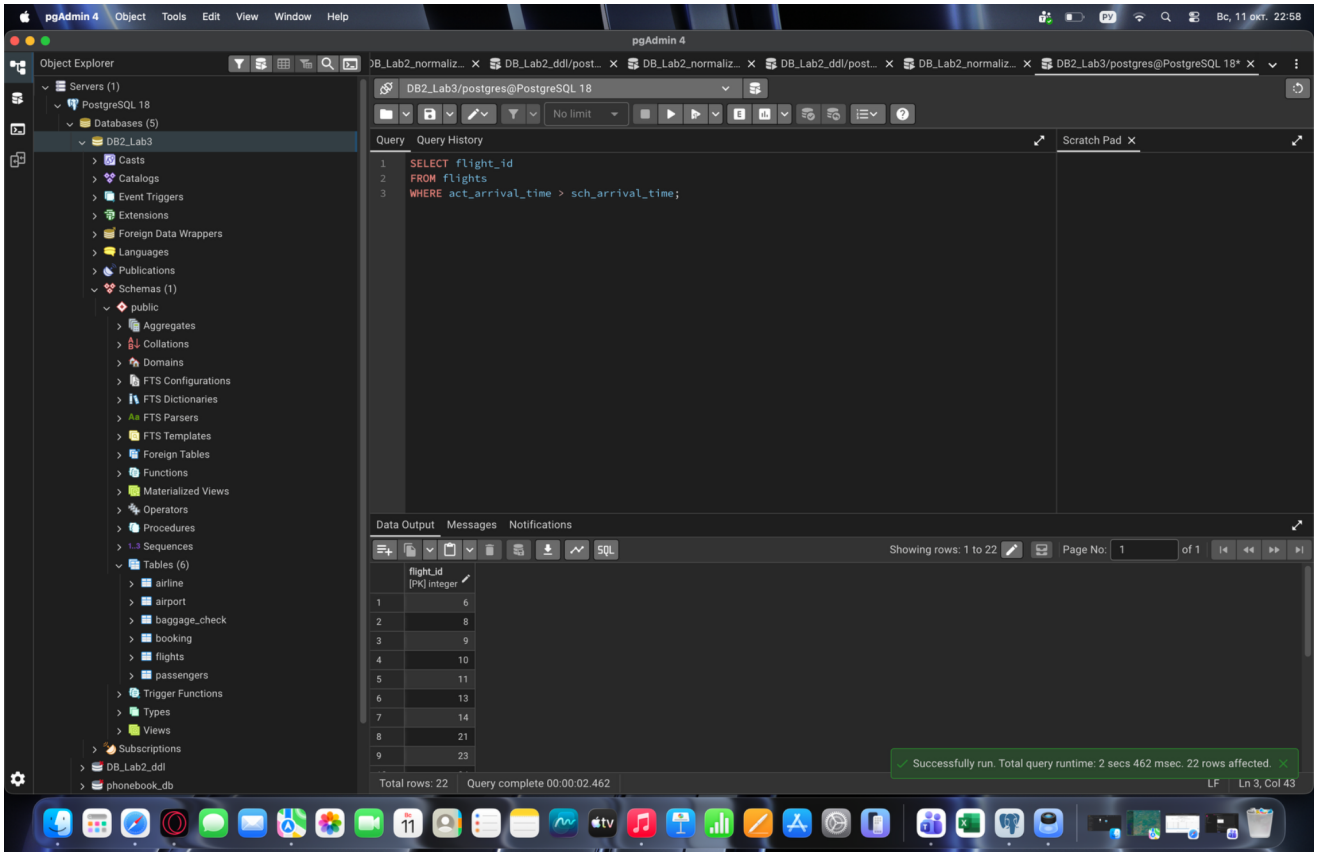
Task 6

Objective: Find flight numbers that were delayed based on actual arrival time.

SQL:

```
SELECT flight_id
FROM flights
WHERE act_arrival_time > sch_arrival_time;
```

Result: 22 delayed flights were found.



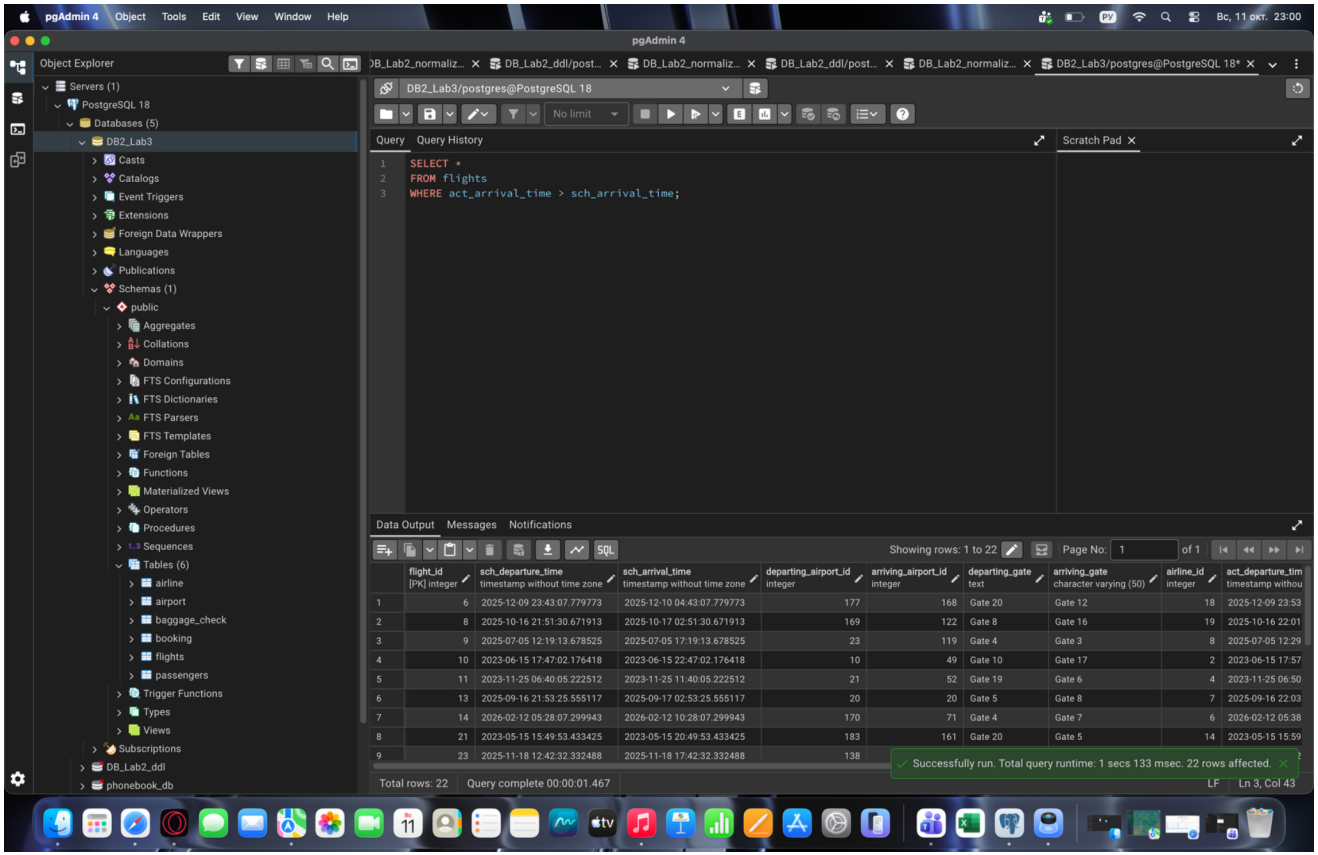
Task 7

Objective: Find flights that arrived later than their scheduled arrival time.

SQL:

```
SELECT *
FROM flights
WHERE act_arrival_time > sch_arrival_time;
```

Result: 22 rows with full information about late-arriving flights were returned.



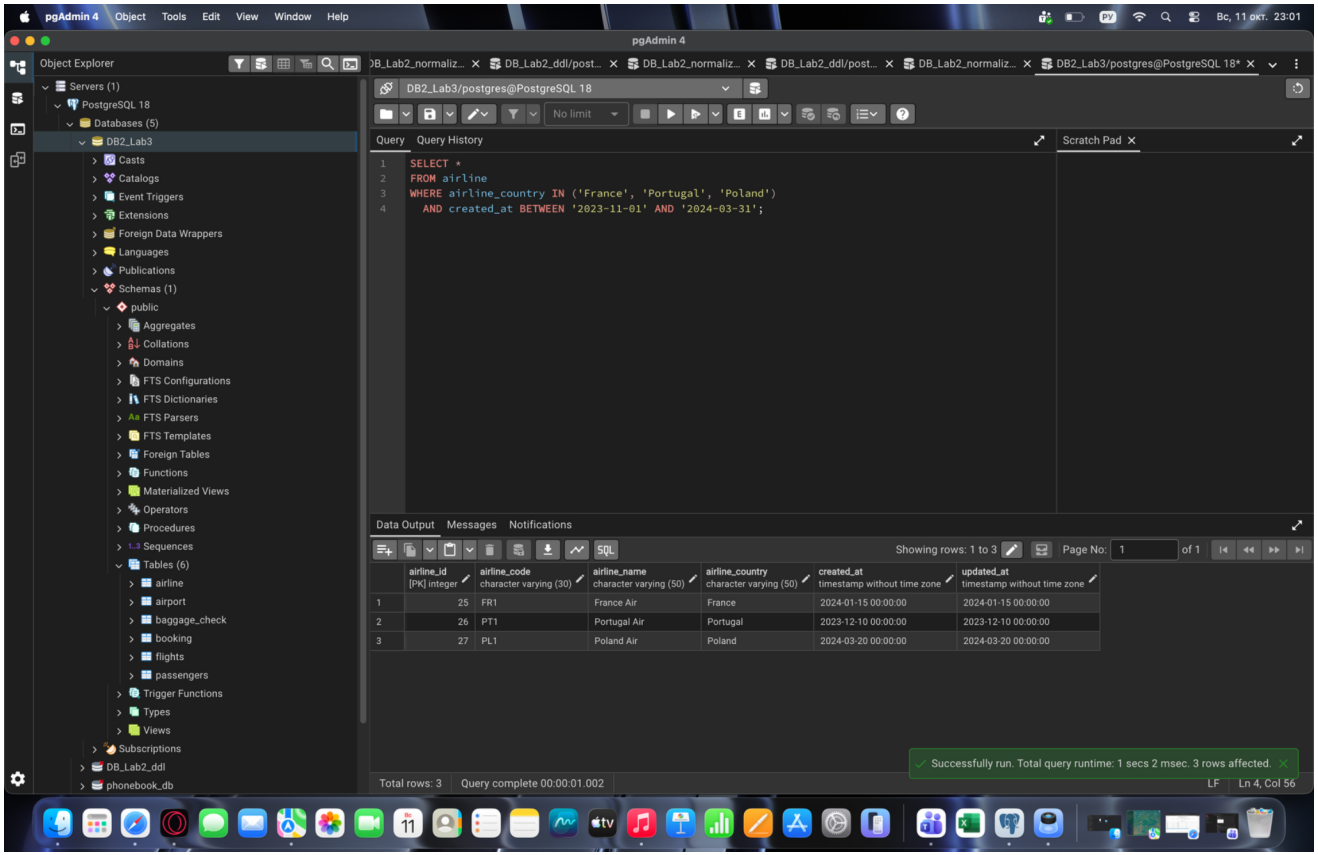
Task 8

Objective: Show airlines from France, Portugal, or Poland created between 2023-11-01 and 2024-03-31.

SQL:

```
SELECT *
FROM airline
WHERE airline_country IN ('France', 'Portugal', 'Poland')
  AND created_at BETWEEN '2023-11-01' AND '2024-03-31';
```

Result: Three matching airlines were returned: France Air, Portugal Air, and Poland Air.



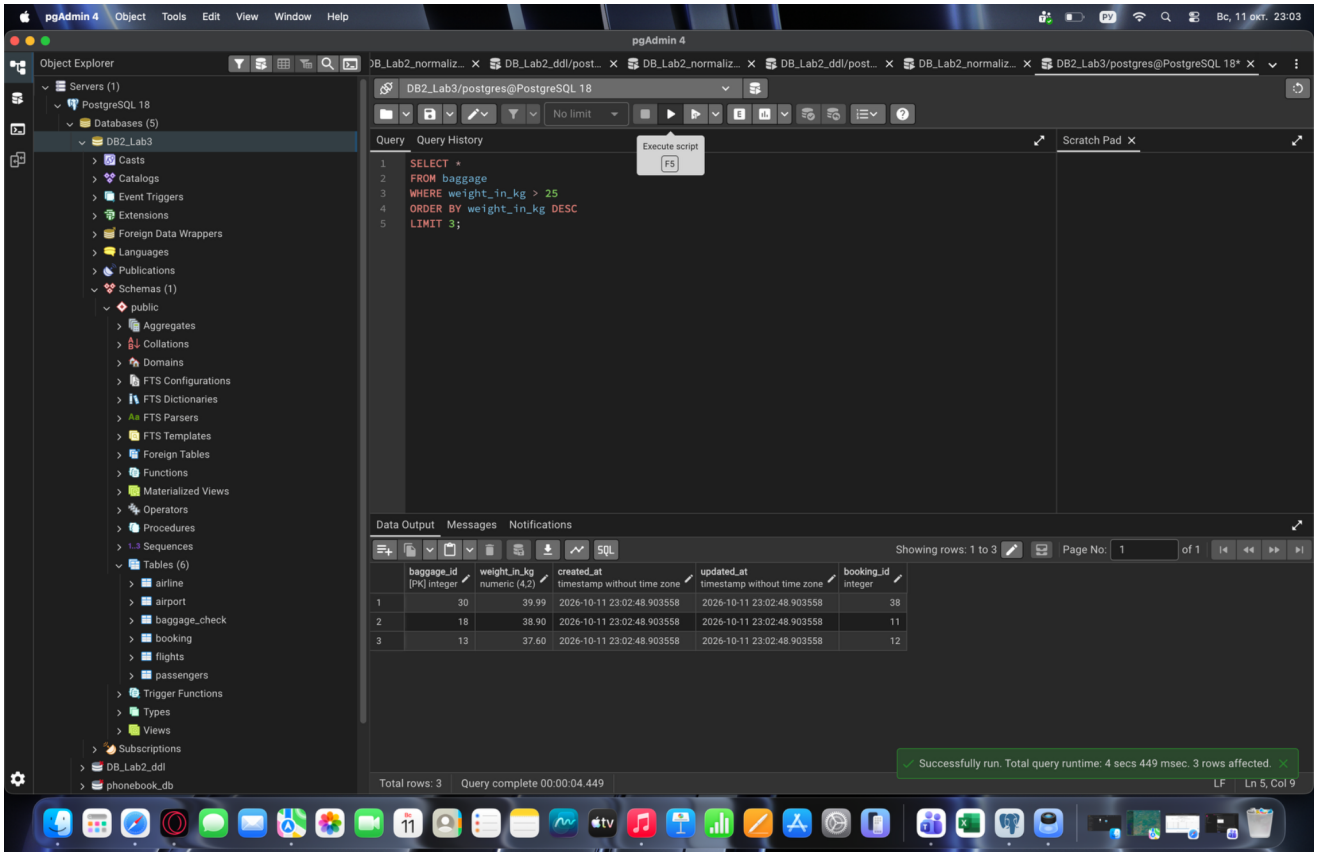
Task 9

Objective: Find the top 3 overweight baggage records with weight greater than 25 kg.

SQL:

```
SELECT *
FROM baggage
WHERE weight_in_kg > 25
ORDER BY weight_in_kg DESC
LIMIT 3;
```

Result: The three highest weights were 39.99 kg, 38.90 kg, and 37.60 kg.



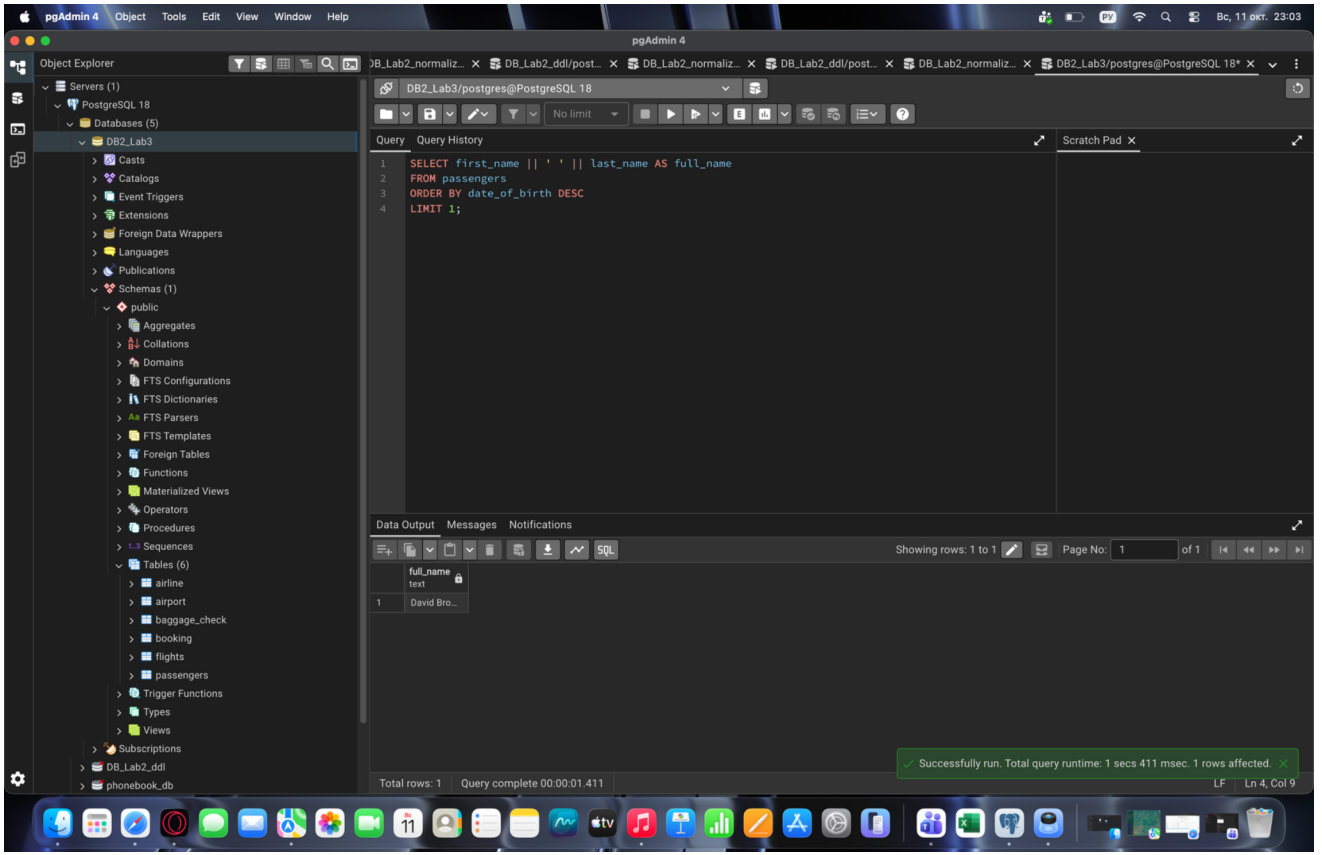
Task 10

Objective: Find the full name of the youngest passenger.

SQL:

```
SELECT first_name || ' ' || last_name AS full_name
FROM passengers
ORDER BY date_of_birth DESC
LIMIT 1;
```

Result: One youngest passenger was returned.



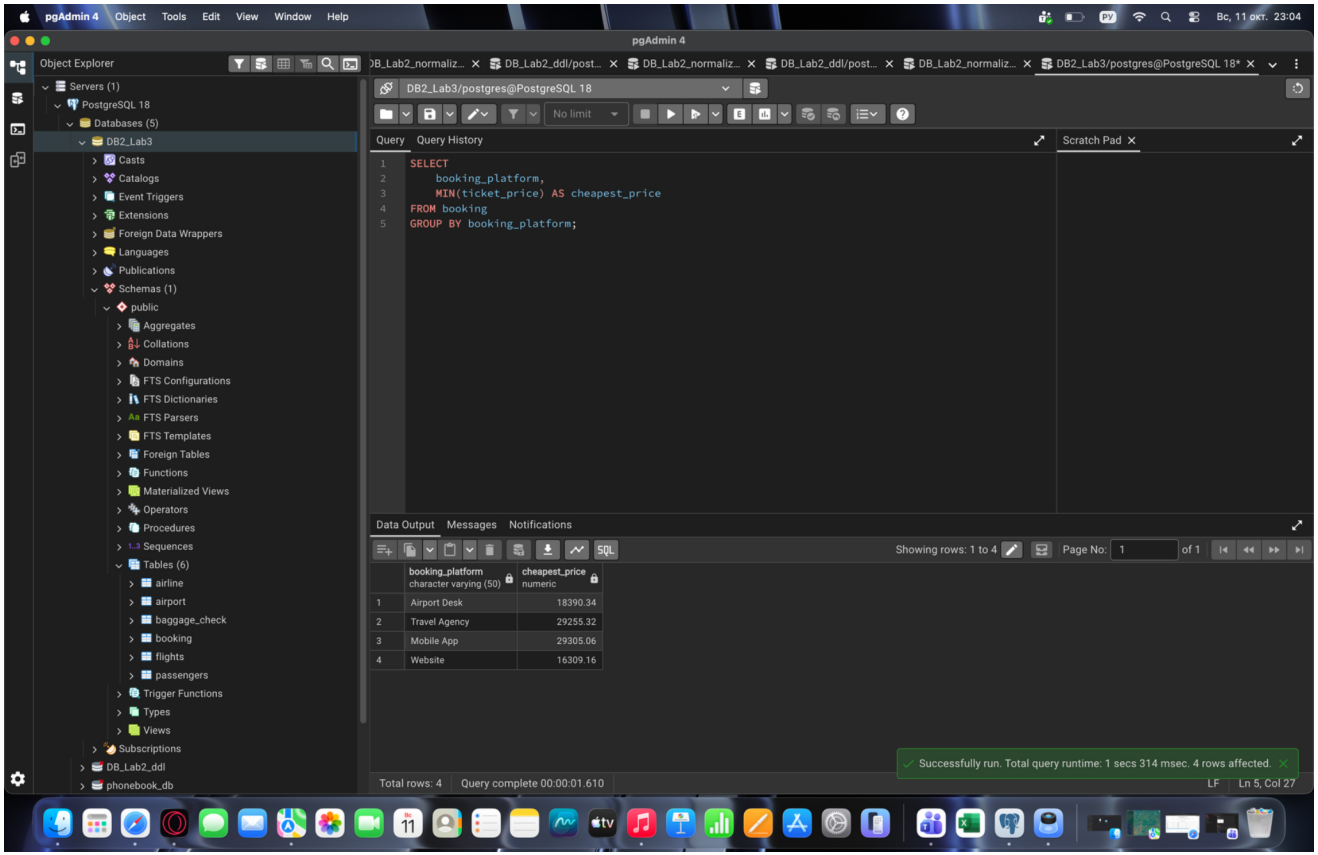
Task 11

Objective: Find the cheapest booking price for each booking platform.

SQL:

```
SELECT
    booking_platform,
    MIN(ticket_price) AS cheapest_price
FROM booking
GROUP BY booking_platform;
```

Result: Minimum booking prices were returned for Airport Desk, Travel Agency, Mobile App, and Website.



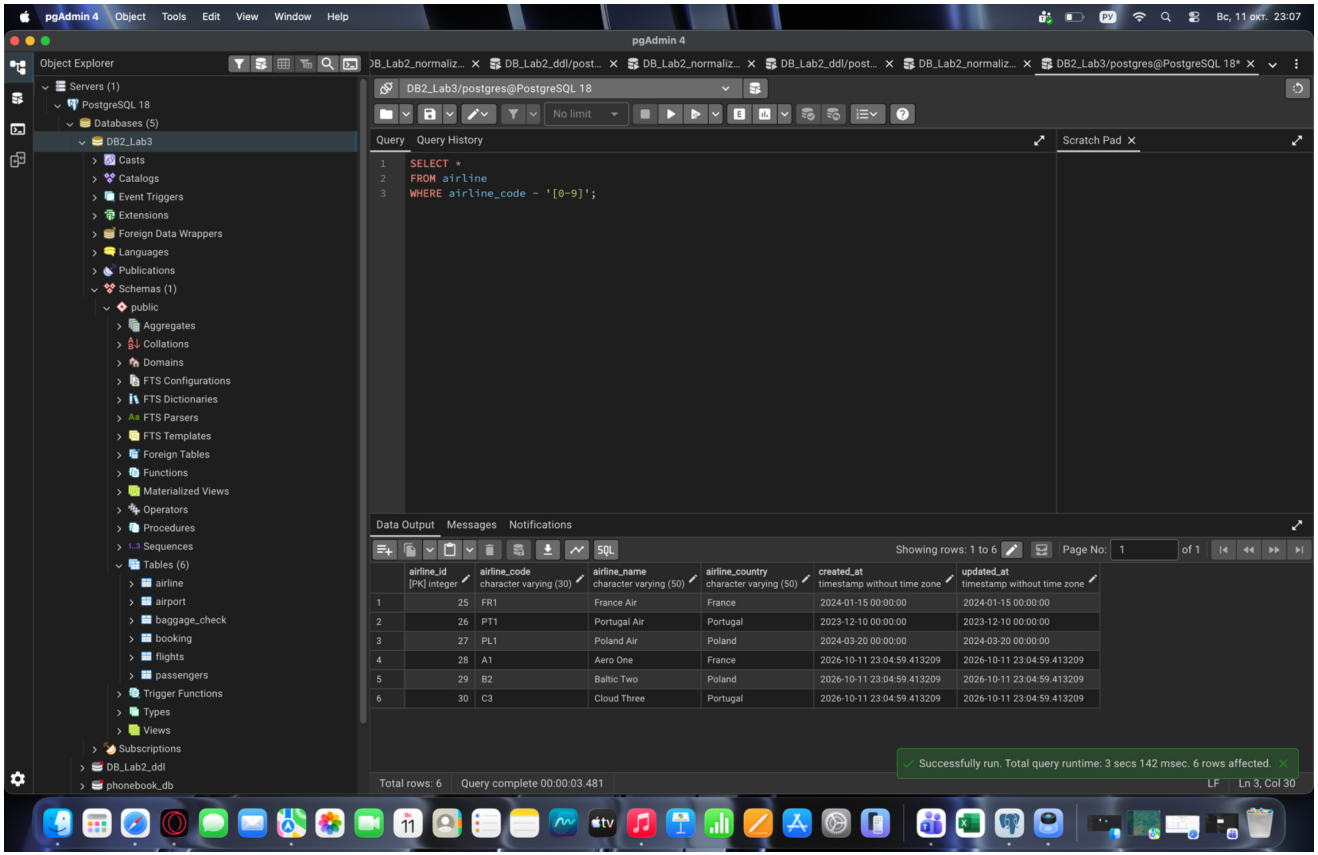
Task 12

Objective: Return airlines whose airline_code contains at least one digit.

SQL:

```
SELECT *
FROM airline
WHERE airline_code ~ '[0-9]';
```

Result: Six airlines with digits in airline_code were returned.



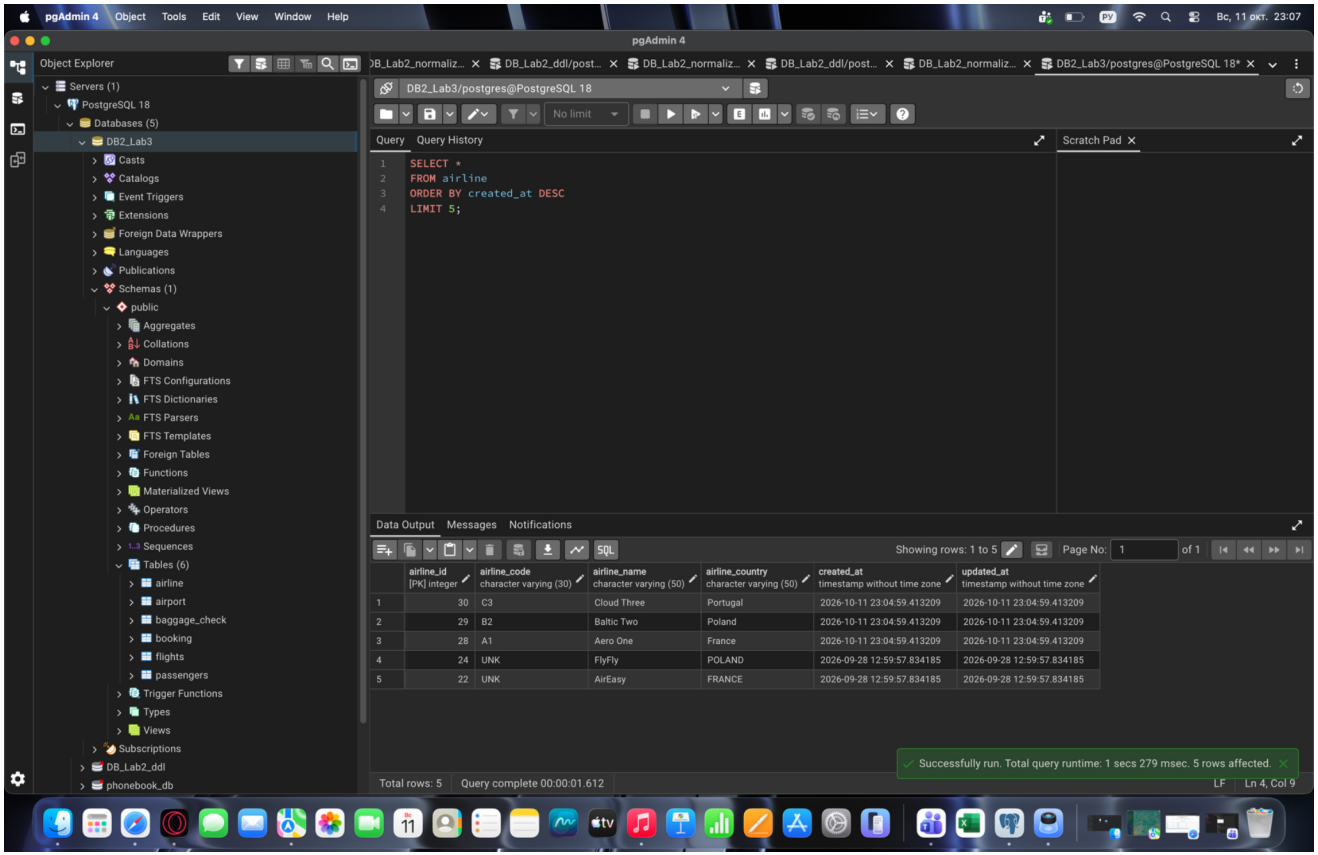
Task 13

Objective: List the top 5 most recently created airlines.

SQL:

```
SELECT *
FROM airline
ORDER BY created_at DESC
LIMIT 5;
```

Result: The five most recently created airlines were returned.



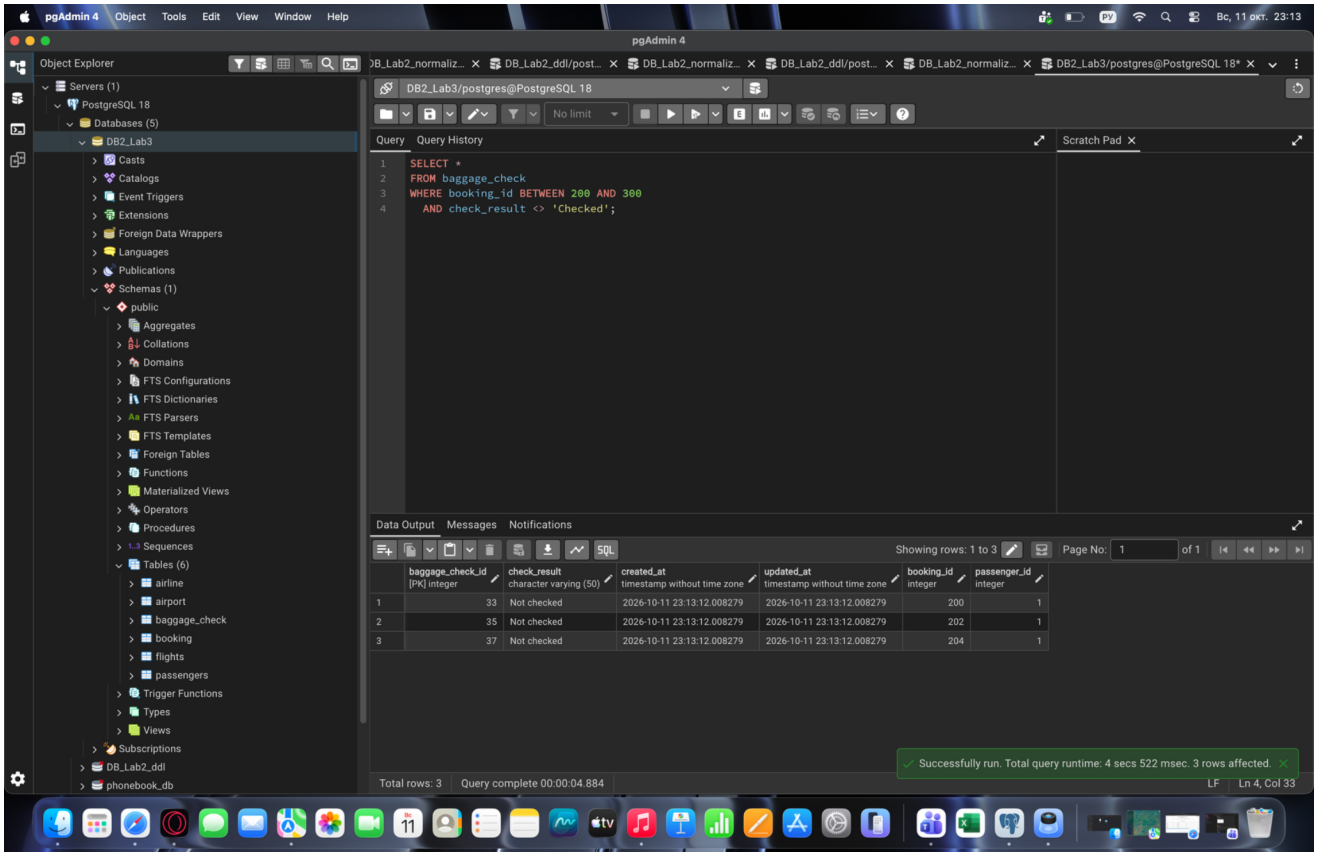
Task 14

Objective: Return baggage_check rows where booking_id is between 200 and 300 inclusive and check_result is not Checked.

SQL:

```
SELECT *
FROM baggage_check
WHERE booking_id BETWEEN 200 AND 300
  AND check_result <> 'Checked';
```

Result: Three matching rows were returned: booking_id 200, 202, and 204 with status Not checked.



Task 15

Objective: Find baggage checks where updated_at is in the same month as created_at but occurs earlier than created_at.

SQL:

```
SELECT *
FROM baggage_check
WHERE DATE_TRUNC('month', updated_at) = DATE_TRUNC('month', created_at)
  AND updated_at < created_at;
```

Result: Three records matching the condition were returned.

